

EE 641: Advance Wireless Technologies

Course Code	EE 641
Course Name	Advance Wireless Technologies
L-T-P-C	3-0-0-3
Prerequisites	EE304, EE503
Intended for	UG and PG
Distribution	Discipline elective for 3rd/ 4th year B.Tech. (CSE and EE)/ MS/ M.Tech./ PhD and free elective for other branches.
Approval	recommended by the 36th BoA (Sep 2020)

Course Modules

- **Introduction**
Challenges of next generation wireless networks. Basics of wireless communications, multipath propagation and fading nature of wireless channel, BER. Performance of multi-antenna wireless systems, precoding and power allocation for multi-user MIMO systems. [7 Hours]
- **Cognitive Radio Systems**
Concepts, challenges of Software Defined Radio (SDR), spectrum-sensing techniques, optimal power allocation, Interference suppression, and robust detection. [5 Lectures]
- **Massive MIMO Systems**
Introduction and challenges. Signal processing with perfect & imperfect channel state information, rate scaling, performance of multi-cell massive MIMO systems and spatial modulation. [7 Lectures]
- **mmWave/THz Wireless Systems**
Introduction, properties and modeling of wireless channels, analog, digital and hybrid processing, sparse processing, channel estimation, optimal precoders and combiners. [6 Lectures]
- **Cooperative Wireless Communication**
Introduction to cooperative communication and cooperation protocols (AF, DF, and SDF). Performance analysis of DF for MIMO and multi-relay wireless systems. [6 Lectures]
- **Non-Orthogonal Multiple-Access (NOMA)**
Introduction, system model and decoding for NOMA systems. Outage probability, opt. performance, average rate, key aspects of uplink and downlink NOMA systems. [6 Lectures]
- **Full-Duplex Wireless Technology** Introduction, self-interference and resulting performance. Optimal signal processing, power allocation and performance of FD Systems. [5 Lectures]
- **Optional topics**
Molecular Communication, Backscatter Communication, Energy Harvesting, Low Power Wide-area Networks (LPWA), Long Range Wireless Transfer (LoRa), D2D, Distributed MIMO, Physical layer caching and Physical layer security.

Text Books

1. Luo, Fa-Long and Zhang, Charlie (Jianzhong), Luo, Fa-Long, **Signal processing for 5G: algorithms and implementations**, Wiley-IEEE Press, 2017.
2. Wei Xiang, Kan Zheng, Xuemin, **5G Mobile Communications**, Springer, 2017

References

1. Wong V., Schober, R., Ng, D., and Wang, L. (Editors), **Key Technologies for 5G Wireless Systems**, Cambridge, 2017.
2. Dahlman, E., Parkvall, S., & Skold, J., **5G NR: The next generation wireless access technology**, Elsevier 2018.
3. A. Chockalingam, B. S. Rajan, **Large MIMO Systems**, Cambridge, 2014.
4. Liu, K., Sadek, A., Su, W., and Kwasinski, A., **Cooperative Communications and Networking**, Cambridge University Press, 2008.
5. Tho Le-Ngoc, Ahmed Masmoudi, **Full-Duplex Wireless Communications Systems**, Springer, 2017.
6. Biglieri, Ezio, Andrea J. Goldsmith, Larry J. Greenstein, Narayan B. Mandayam and Herve Vincent, **Principles of Cognitive**, Cambridge, 2012.
7. N. Farsad, H. B. Yilmaz, A. Eckford, C. Chae and W. Guo, *A Comprehensive Survey of Recent Advancements in Molecular Communication*, in **IEEE Communications Surveys & Tutorials**, vol. 18, no. 3, pp. 1887-1919, 2016.